

Chung-Yen (Matt) Chiang

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Professional Experience

Amazon - Connectivity Team

Apr 2026 – Present

Software Development Engineer II (L5)

- Lifted end-to-end Wi-Fi throughput **2.36x** (10.59 to 25 Mbps) on **FreeRTOS** via cross-layer profiling; resolved IMI memory bottleneck (SDIO), TCP shared-lock and mutex contention (kernel), and application memory contention, plus CPU overclock tuning.
- Architected runtime **dynamic NVRAM/CLM loading** from flash, enabling a single firmware image to serve multiple SKUs and regulatory regions without per-region rebuild.
- Led **Wi-Fi/BLE bring-up** on next-gen ASIC platforms; integrated wireless chipset BSPs/SDKs into production FreeRTOS firmware for NPI programs.

Amazon - Connectivity Team

Aug 2023 – Mar 2026

Software Development Engineer I (L4)

- Built a **modular BLE setup subsystem** from scratch with an **E2E encryption channel** and dynamic country-code configuration, enabling global SKU deployment from a single firmware base.
- Optimized DHCP reconnection logic, reducing reconnection latency by **76%** under unstable network conditions.
- Tuned **WMM QoS (EDCA)** parameters to prioritize latency-sensitive traffic, improving p99 latency for streaming services by **24%**.
- Implemented Wi-Fi power-saving and wake-up mechanisms, extending battery-operated IoT device standby from **40 to 65 days**.

Amazon - Connectivity Team

Sep 2022 – Jul 2023

Software Development Engineer Intern

- Improved priority-based connection logic, raising connection success rate across IoT ecosystems.
- Developed automated tools for interoperability testing across **60+ commercial router models**.

Academia Sinica - Institute of Information Science

Jul 2022 – Aug 2022

Research Intern (Advised by Prof. Meng Chang Chen)

- Researched hardware-level evasion techniques (CPU/registry checks) to improve detection accuracy of analysis environments.

Research & Projects

Secure Channels for Smart Manufacturing Systems in Cloud Environments (Advised by Prof. Nai Wei Lo)

Jul 2021 – Feb 2022

1st Author, CISC 2022, Recipient, MOST Undergraduate Research Scholarship

- Researched and implemented **Site-to-Site IPsec** secure channels for industrial communication between local gateways and AWS VPC.

Technical Skills

Core Domains	Embedded Systems, Wi-Fi & BLE, RTOS (FreeRTOS), BSP/SDK Integration, Cross-Layer Performance Profiling, Deep Packet Inspection
Programming	C, C++, Python, Shell Scripting
Tools	Linux, GDB / JTAG, perf, Wireshark, tcpdump, iperf3, Git
Languages	English (Professional), Mandarin (Native)

Education

National Taiwan University of Science and Technology

2019 – 2023

B.B.A. in Information Management

GPA: 4.01/4.30

Academic Excellence Award - Top 2 of class for three consecutive semesters